

AI USAGE DISCLOSURE FORM

Samsung PRISM Generative AI Hackathon (3rd Edition 2026–27)

1. Team Details

Team Name: VIT_CodeNexus_1

Project / Product Name: Agentic Code Intelligence — High-Precision Dense Code Retrieval Engine (Theme 01)

Organization / Institution (if any): Vellore Institute of Technology

Submission Date: 2026-09-30

2. AI Usage Declaration

Did your team use any Artificial Intelligence (AI) in developing this project? **YES**

Declaration Summary: Artificial intelligence tools (Claude and Google Antigravity) were utilized as assistive pair-programming, exploratory research, and documentation aids during project development. All retrieval decisions, benchmark compliance, evaluation verification, and architectural validation were directed, tested, and validated by the CodeNexus team members. All reported metrics come from actual CPU evaluation runs and were not generated by AI.

3. Purpose of AI Usage (Brief Details)

Idea generation / brainstorming: Explored candidate embedding models (E5, Jina Code, EmbeddingGemma) and task-specific instruction prompt strategies compatible with MTEB AppsRetrieval and CPU execution bounds.

Code generation or assistance: Implemented the PrePostPipelineEncoder(AbsEncoder) wrapper, FAISS IndexFlatIP retrieval index, candidate comparison runner, and interactive CLI demo.

UI / UX design: N/A (Project is a backend retrieval pipeline and command-line search prototype; no graphical user interface developed).

Content creation: Generated technical documentation (README.md), slide deck content matching the official Samsung template (submission/PPT_CONTENT.md), and submission verification checklists. Claude was also used to refine the layout, wording, and consistency of the final presentation (submission/CodeNexus_PRISM_Theme01_Final.pptx and .pdf) and to review this disclosure form; project metrics and technical facts were not changed.

Data analysis: Analyzed Experiment 1 internal validation metrics across 200 queries and extracted official full-test evaluation metrics from MTEB output JSON.

Testing / debugging: Implemented automated test suites (test_baseline.py, test_demo.py), resolved MTEB 2.21 datetime JSON serialization issues, and handled HuggingFace gated repository fallback handling.

Other: N/A

4. Feature Origin Classification

Classification of all 7 key project features according to origin, tools used, prompt purpose, outputs, and human modifications:

Feature 1: Baseline PrePostPipelineEncoder & FAISS DenseIndex Utility

1. **Feature Name:** Baseline PrePostPipelineEncoder & FAISS DenseIndex Utility

2. **Classification:** Both (Human-Architected + AI-Assisted)

3. **Description Details:**

- **AI Tools / Platform Used:** Claude & Google Antigravity
- **Prompt / Intent:** Implement PrePostPipelineEncoder inheriting from MTEB SentenceTransformerEncoderWrapper / AbsEncoder and create an isolated FAISS IndexFlatIP dense indexing utility with NumPy fallback.
- **Output Summary:** src/encoder.py, src/index/dense_index.py, eval/run_mteb.py
- **Human Modification / Verification:** Reviewed inheritance compliance with MTEB 2.21, verified unit vector L2 normalization math, and confirmed baseline evaluation execution on CPU.

Feature 2: Robust MTEB JSON Serialization & Datetime Handling

1. **Feature Name:** Robust MTEB JSON Serialization & Datetime Handling

2. **Classification:** Both (Human-Architected + AI-Assisted)

3. **Description Details:**

- **AI Tools / Platform Used:** Google Antigravity
- **Prompt / Intent:** Debug and resolve TypeError: Object of type datetime is not JSON serializable occurring during json.dump() of MTEB TaskResult.to_dict().
- **Output Summary:** Added default=str serialization handler in eval/run_mteb.py.
- **Human Modification / Verification:** Validated that saved JSON artifacts are strictly valid JSON and retain all metric keys without corruption.

Feature 3: Candidate Model Comparison Runner (Validation Slice)

1. **Feature Name:** Candidate Model Comparison Runner (Validation Slice)

2. **Classification:** Both (Human-Architected + AI-Assisted)

3. **Description Details:**

- **AI Tools / Platform Used:** Claude & Google Antigravity
- **Prompt / Intent:** Build a deterministic validation harness (200 held-out train queries vs. 1,500 candidate documents) to benchmark intfloat/e5-base-v2, jina-embeddings-v2-base-code, and google/embeddinggemma-300m under identical flat cosine retrieval conditions.
- **Output Summary:** configs/experiment_1_models.yaml, eval/run_experiment.py, eval/results/experiment_1_*.json
- **Human Modification / Verification:** Configured fixed random seed (42), reviewed per-model asymmetric query/passage prompts, capped sequence lengths to 512 for fair CPU evaluation, and selected embeddinggemma-300m as the decisive winner.

Feature 4: EmbeddingGemma Integration with Automated Mirror Fallback

1. **Feature Name:** EmbeddingGemma Integration with Automated Mirror Fallback

2. Classification: **Both (Human-Architected + AI-Assisted)**

3. Description Details:

- **AI Tools / Platform Used:** Google Antigravity
- **Prompt / Intent:** Enable google/embeddinggemma-300m to run where the gated model repository cannot be accessed (e.g., unauthenticated judge machines) by falling back to the identical-weight RedHatAI/embeddinggemma-300m mirror, and attach official task prompts.
- **Output Summary:** Mirror fallback and prompt mapping logic in src/encoder.py.
- **Human Modification / Verification:** Verified weight and architecture identity, tested fallback behavior under synthetic 403 errors, and confirmed 768-dimensional normalized output.

Feature 5: Judge-Ready CLI Code Search Prototype

1. Feature Name: Judge-Ready CLI Code Search Prototype

2. Classification: **Both (Human-Architected + AI-Assisted)**

3. Description Details:

- **AI Tools / Platform Used:** Google Antigravity
- **Prompt / Intent:** Create a standalone CLI tool (demo.py) that runs locally on CPU once model weights are cached, indexes a curated 15-algorithm code corpus (data/demo_corpus.json), accepts natural language queries, and outputs ranked code snippets with latency.
- **Output Summary:** demo.py, data/demo_corpus.json
- **Human Modification / Verification:** Curated representative algorithms covering searching, sorting, graph traversal, and dynamic programming (e.g., binary search, merge sort), verified score rankings, and silenced library warnings for clean CLI output.

Feature 6: Comprehensive Pytest Test Suite

1. Feature Name: Comprehensive Pytest Test Suite

2. Classification: **Both (Human-Architected + AI-Assisted)**

3. Description Details:

- **AI Tools / Platform Used:** Google Antigravity
- **Prompt / Intent:** Create automated test cases covering encoder AbsEncoder compliance, vector unit length, FAISS exact search, demo corpus integrity, and CLI subprocess execution.
- **Output Summary:** tests/test_baseline.py, tests/test_demo.py, pytest.ini
- **Human Modification / Verification:** Verified that all 9 tests pass with 100% pass rate in local environment.

Feature 7: Containerization & Technical Submission Documentation

1. Feature Name: Containerization & Technical Submission Documentation

2. Classification: **Both (Human-Architected + AI-Assisted)**

3. Description Details:

- **AI Tools / Platform Used:** Claude & Google Antigravity
- **Prompt / Intent:** Generate a minimal python:3.10-slim container definition, detailed project documentation (README.md), slide deck markdown (submission/PPT_CONTENT.md), final submission checklists, and a layout/wording refinement of the final presentation.
- **Output Summary:** Dockerfile, .dockerignore, README.md, submission/PPT_CONTENT.md, submission/SUBMISSION_CHECKLIST.md, submission/CodeNexus_PRISM_Theme01_Final.pptx, submission/CodeNexus_PRISM_Theme01_Final.pdf

- **Human Modification / Verification:** Verified all documented commands, verified metric numbers against official test output, ensured no unverified or out-of-scope features were claimed, and checked that the refined presentation kept all metrics unchanged and labelled P1/P2 items as Future Work.

5. Ethical & Compliance Confirmation

AI usage complies with guidelines and policies: **YES**

No proprietary or copyrighted data misused: **I AGREE**

6. Declaration & Sign-Off

Name of Team Representative: Mithun Venkatesan

Role: Team Lead (CodeNexus)

Signature: Mithun Venkatesan

Date: 2026-09-30